

Christopher P. Ravosa

Boston, MA (Open to Remote) | christopher.ravosa99@gmail.com

Website: <https://www.chrisravosa.com/> | **LinkedIn:** <https://www.linkedin.com/in/christopherravosa/>

SUMMARY

Software engineer focused on backend systems, real-time distributed services, and high-throughput data processing. Experienced building production APIs, scalable data pipelines, and reliable systems under load.

SKILLS

Languages:	C#, Python, TypeScript, JavaScript, C++, Java
Systems:	Distributed Systems, Real-Time Systems, TCP Networking, REST APIs
Data & Infrastructure:	MongoDB, NoSQL Modeling, Aggregation Pipelines, GCP, AWS
Tools:	Node.js, React Native, Git, CI/CD

PROFESSIONAL EXPERIENCE

Senior Software Engineer - Moku

Mar 2025 - Present

- ❖ Built systems to run and manage large-scale parallel simulations for training and evaluating autonomous agents
- ❖ Implemented a runtime configuration system for defining simulation environments and agent behavior, reducing experimental iteration from weeks to hours
- ❖ Developed a deterministic event and state recording pipeline for debugging, auditing, and replay of system behavior
- ❖ Designed modular runtime systems that could be enabled or disabled based on execution context, improving flexibility and debugging
- ❖ Created internal tooling including structured logging, debugging dashboards, and configuration management systems to improve developer velocity

Associate Software Engineer - Major League Baseball

Aug 2023 - Mar 2025

- ❖ Developed TCP-based matchmaking services and backend APIs supporting real-time user interactions
- ❖ Built backend systems to handle high-throughput requests and maintain performance under peak load conditions
- ❖ Designed and executed automated stress testing frameworks for backend services, identifying memory leaks and data integrity issues and improving system reliability under sustained load
- ❖ Implemented data processing and leaderboard systems handling large player datasets, leveraging MongoDB aggregation pipelines to efficiently query and rank millions of records
- ❖ Contributed to content management system enabling dynamic configuration of live service content without direct database access
- ❖ Integrated external data sources including MLB StatCast APIs, transforming high-volume, real-world data into interactive systems
- ❖ Participated in on-call rotations, diagnosing and resolving production incidents in live systems, including recovery of backend services during outages
- ❖ Built internal tools and pipelines for asset distribution via CDN (GCP), enabling dynamic content delivery while minimizing client footprint

Associate Technical Designer - Naughty Dog

Jul 2022 - Jun 2023

- ❖ Implemented networked systems using a host-authoritative architecture, ensuring state consistency across distributed clients
- ❖ Built synchronization logic resilient to latency, packet loss, and unreliable network conditions
- ❖ Designed and tested systems to prevent state divergence and maintain consistency in real-time environments

EDUCATION

- ❖ **MSc in Software Development - Concentration in Mobile Computing**
Marist College - *Sept. 2020 to May 2022*